1. Information Representation

1.1. Data Representation

- The two fundamental characteristics of any number system are:
  - **A base:** The number of different digits that a system can use to represent numbers
  - **Place value:** The specific-value of a digit based on its position within a number
- **Denary** - Base 10
- **Binary Systems** - Base 2
  - Possible bits (binary digits): 0 and 1
  - All data and characters are represented in binary

128	64	32	16	8	4	2	1
0	0	0	0	0	0	0	0

- For example, 65 in binary is *01000001*

- **Denary vs. Binary prefixes:**

Denary Prefix	Factor Value	Binary Prefix	Factor Value
kilo- (k)	$\times 10^3$	kibi- (Ki)	$\times 2^{10}$
mega- (M)	$\times 10^6$	mebi- (Mi)	$\times 2^{20}$
giga- (G)	$\times 10^9$	gibi- (Gi)	$\times 2^{30}$
tera- (T)	$\times 10^{12}$	tebi- (Ti)	$\times 2^{40}$

- **Binary Coded Decimal (BCD)**
  - Binary representation where each positive denary digit is represented by a sequence of 4 bits (nibble).
  - Only certain digits are converted to BCD because particular digits represent a digit greater than 9.
  - For example, to represent 429 in BCD:
  - Convert each individual digit of the number to its binary equivalents -
    1. *4 = 0100*
    2. *2 = 0010*
    3. *9 = 1001*
  - Concatenate the 3 nibbles (4-bit group) in order to produce BCD: *0100 0010 1001*
- **Practical applications**
  - A string of digits on any electronic device displaying numbers (like in calculators)
  - Accurately measuring decimal fractions
  - Electronically coding denary numbers
- **Two's Complement**
  - We can represent a negative number in binary by making the most significant bit (MSB) a sign bit, which indicates whether the number is positive or negative.
  - **Converting from negative denary to binary two's complement (example -42):**
    1. Find the binary equivalent of the denary number (ignoring the negative sign) | 42 = 101010
    2. Add extra 0 bits before the MSB, to format the binary number to 8 bits | 00101010
    3. Convert binary number to one's complement (flip the bits) | 11010101
    4. Convert binary number to two's complement (add 1) |1010101 + 1 = 11010110
  - **Converting binary two's complement into denary (example 11010110):**
    1. Flip all the bits | 00101001
    2. Add 1 | 00101010
    3. Convert binary to denary and put a negative sign) | -42
- Maximum positive number in 8 bits: 127
- Maximum negative number in 8 bits: -128
- **Hexadecimal Systems - Base 16:**
  - Possible digits: 0 to 9 and A to F, where A to F represent denary digits 10 to 15
  - Practical applications:
    - Defining colours in HTML
    - Defining Media Access Control (MAC) addresses
    - Assembly languages and machine code
    - Debugging via memory dumps
  - Example - A5 in Denary = (16×10) + (1×5) *= 165*
  - Example - 65 in Hexadecimal = 65÷16 = 4 | Remainder 1 | *∴ = 41*
- **Character Sets:**
  - A character set generally includes upper & lower case letters, number digits, punctuation marks, and other characters.
  - Character sets use different binary representations for each character via character encoding

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- **Character Encoding Standards:**

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ASCII	Extended ASCII	Unicode
Only the English alphabet can be represented	ASCII's extension - Also includes most European languages' alphabets	Superset for ASCII & extended ASCII - recognized by various global languages
Each character encoding takes up 7 bits, hence 128 possible characters	ASCII extended to 8 bits, hence 256 possible characters	It has an excellent range of characters, using 2 or 4 bytes per character
Smaller storage space	2 or 4 times more storage space per character	

1.2. Multimedia - Graphics, Sound

- Bitmap Images
  - *Definition:* Bitmap images are created by assigning a solid color to each pixel using bit patterns.
  - *Encoding:*
    - Bit patterns are generated by considering each grid row as a series of binary color codes corresponding to each pixel's color.
    - These bit patterns are 'mapped' onto the main memory.
  - *Pixels:* The smallest picture element whose color can be accurately represented by binary code.
  - *File Header:* Bitmap images contain a file header with metadata, including image size, number of colors, etc.

- Image Resolution
  - *Definition:* the number of pixels that make up an image. Example:- 4096x3192 pixels.
  - *Effect:* Higher resolution results in sharper, more detailed images.

- Screen Resolution
  - *Definition:* The number of pixels that can be viewed horizontally and vertically on a device's screen.
  - *Calculation:* $Number\ of\ pixels = width \times height$ (e.g., 1680 × 1080 pixels).

- Colour depth
  - *Definition:* The number of bits used to represent the colour of a single pixel.
  - *Calculation:* An image with n bits has 2^n colours per pixel (e.g., a 16-color bitmap has 4 bits per pixel because $2^4 = 16$)
  - *Effect:* Increasing color depth improves color quality but also increases file size.
  - *File Size Calculation:*
    - File Size = Number of Pixels × Color Depth
    - Convert bits to bytes by dividing by eight if necessary.
  - *Applications:* Scanned images and general computer usage, where small file sizes and easy manipulation are beneficial.

- Vector Graphics
  - *Definition:* Made up of drawing objects.
  - *Drawing Objects:* Mathematically defined constructs (e.g., rectangles, lines, circles).
  - *Drawing List:* A set of commands defining the vector (e.g., points, lines, shapes, and their attributes that define the image.)
  - *Properties:* Basic geometric data that determine the shape and appearance of each object.
  - *Encoding:* Data is encoded using mathematical formulas to generate properties for drawing lines and curves to create the image.
  - *Scalability:* Objects can be resized without losing quality, unlike bitmaps.
  - *Applications:* Company logos and other graphics that require scaling.

- Sound
  - *Analogue vs Digital:*
    - Analogue data consists of continuous electrical signals.
    - Digital data consists of discrete electrical signals.
  - *Sound Signals:* Vibrations through a medium, inherently analogue due to infinite detail.
  - *Conversion:*
    - Analogue signals are converted to digital signals by sampling.
    - The sound wave's amplitude is sampled at set time intervals.
    - These samples are encoded as a binary number sequence, providing a digital representation of the sound wave

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- Sampling Rate
  - *Definition:* Number of samples taken per unit of time.
  - *Effect:* Increasing the sampling rate improves the accuracy of the digitized sound wave representation but increases file size.

- Sampling Resolution
  - *Definition:* Number of bits used to encode each sample.
  - *Effect:* Increasing sampling resolution improves the accuracy of digitized sound waves but increases file size.

- Bit Rate
  - *Definition:* Number of bits used to store 1 second of sound.
  - *Calculation:* $Bit\ Rate = Sampling\ Rate * Sampling\ Resolution$
  - *Length of Sound:* Measured in seconds.

1.3. Compression

- **Definition:** Compression is the process of reducing file size without significant loss in quality, resulting in:
  - Reduced time needed to search for data.
  - Faster transfer of compressed files, using less bandwidth than uncompressed files.

- Lossless Compression
  - *Definition:* A type of compression that allows original data to be perfectly reconstructed from a compressed file by utilizing some form of replacement.
  - *Examples:* Bitmap (.bmp), vector graphic (.svg), .png images, text file compression, database records.
  - *Run-Length Encoding (RLE):*
    - `Definition`: A form of lossless compression used for compressing text files and bitmap images.
    - `Mechanism`: Reduces file size by encoding sequences of adjacent, identical elements (characters in text files and pixels in bitmap images) into two values: run count and run value.
    - `Example`: RLE of a bitmap image.

- Lossy Compression
  - *Definition:* A type of compression that irreversibly eliminates unnecessary data.
  - *Effect:* File accuracy/quality is lower than with lossless compression, but file size is significantly reduced (often to about 10% of the lossless size).
  - *Examples:* Sound files (.mp3), .jpeg images.
  - *Mechanism in Sound Files:*
    - `Perceptual Coding`: Removes parts of the sound that are less audible or discernible to human hearing, as used in .mp3 compression.

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